



3303 - Pinpointing Early Supermassive Black Holes and Star Formation with JWST and Chandra

Cycle: 2, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

With this proposal we aim to study archival data of $z > 5$ candidate JWST Star-Forming-Galaxies and Early QSO from the COSMOS-Webb, CEERS and PEARLS in conjunction with deep Chandra archival data to probe the early Universe by means of stacking analysis. Our goal is to determine a) constrain the high- z Star Formation Rate (SFR)-L_X efficiency by deriving Chandra X-ray luminosities and SFRs obtained by JWST; b) the abundance of AGN-like X-ray sources at $z > 5$ in order to shed new light on early SMBH growth and c) compute the space density and X-ray throughput of early AGN candidates and compare them with predictions of SMBH seeding models.

OBSERVING DESCRIPTION

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